15. Develop an Android application implementing Runtime Permissions and security handling in AndroidManifest.xml

Project Name:PermissionDemo

Add Permission in AndroidManifest.xml.
Open:
AndroidManifest.xml
Add below code lines above: <application> tag
<uses-permission android:name="android.permission.CAMERA"/>
<uses-feature android:name="android.hardware.camera"
android:required="false" />

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.CAMERA"/>
    <uses-feature android:name="android.hardware.camera"
        android:required="false" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <Button
        android:id="@+id/btnPermission"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Request Camera Permission"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.permissiondemo;


import android.content.pm.PackageManager;

import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


    Button btnPermission;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);


        btnPermission = findViewById(R.id.btnPermission);
```

```
    btnPermission.setOnClickListener(v -> {
        checkPermission();
    });
}

private void checkPermission() {

    // Check if Camera permission is already granted
    if (ContextCompat.checkSelfPermission(
            this,
            android.Manifest.permission.CAMERA
    ) == PackageManager.PERMISSION_GRANTED) {

        Toast.makeText(
                this,
                "Permission Already Granted",
                Toast.LENGTH_SHORT
        ).show();

    } else {

        // Request Camera permission
        ActivityCompat.requestPermissions(
                this,
                new String[]{
                        android.Manifest.permission.CAMERA
```

```
                },
                1
        );
    }
}

// Permission Result
@Override
public void onRequestPermissionsResult(
        int requestCode,
        String[] permissions,
        int[] grantResults
) {
    super.onRequestPermissionsResult(
            requestCode,
            permissions,
            grantResults
    );

    if (requestCode == 1) {

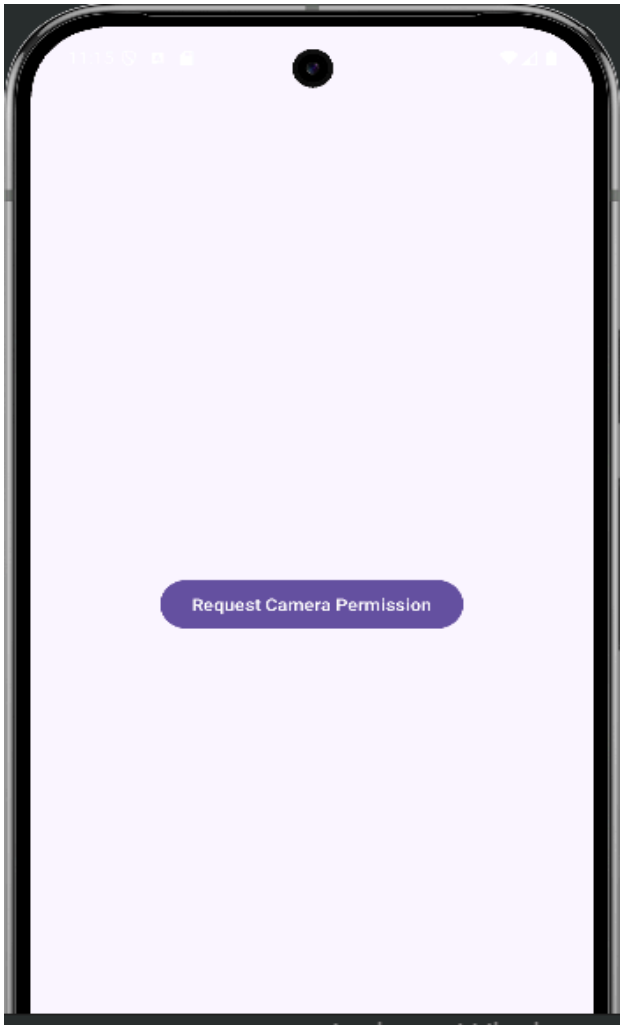
        if (grantResults.length > 0
                && grantResults[0] == PackageManager.PERMISSION_GRANTED) {

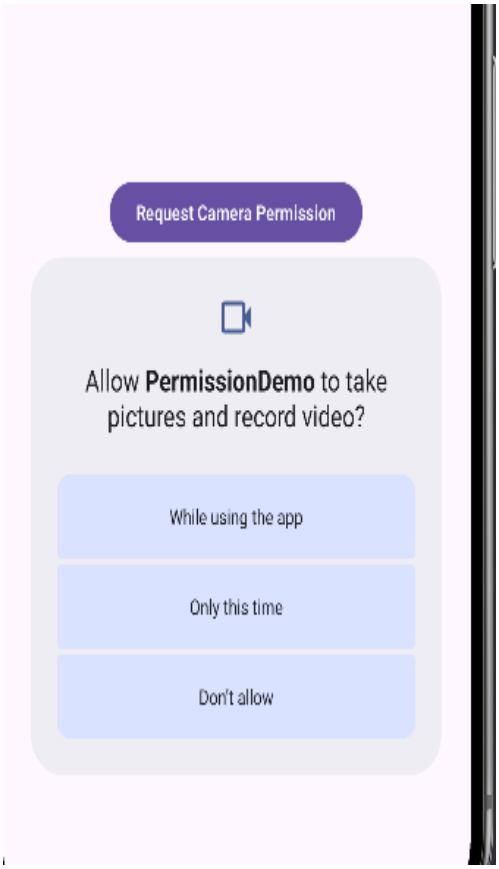
            Toast.makeText(
                    this,
```

```
                        "Permission Granted",
                        Toast.LENGTH_SHORT
                ).show();


            } else {


                Toast.makeText(
                        this,
                        "Permission Denied",
                        Toast.LENGTH_SHORT
                ).show();
            }
        }
    }
}
```


Request Camera Permission


Request Camera Permission
Allow PermissionDemo to take pictures and record video?
While using the app
Only this time
Don't allow